T03 – Thermodynamic Laws and Processes

REGIONAL WEBSITE RELEASE 1.0

Name/team: _____ Date: _____

24 questions | 60 points

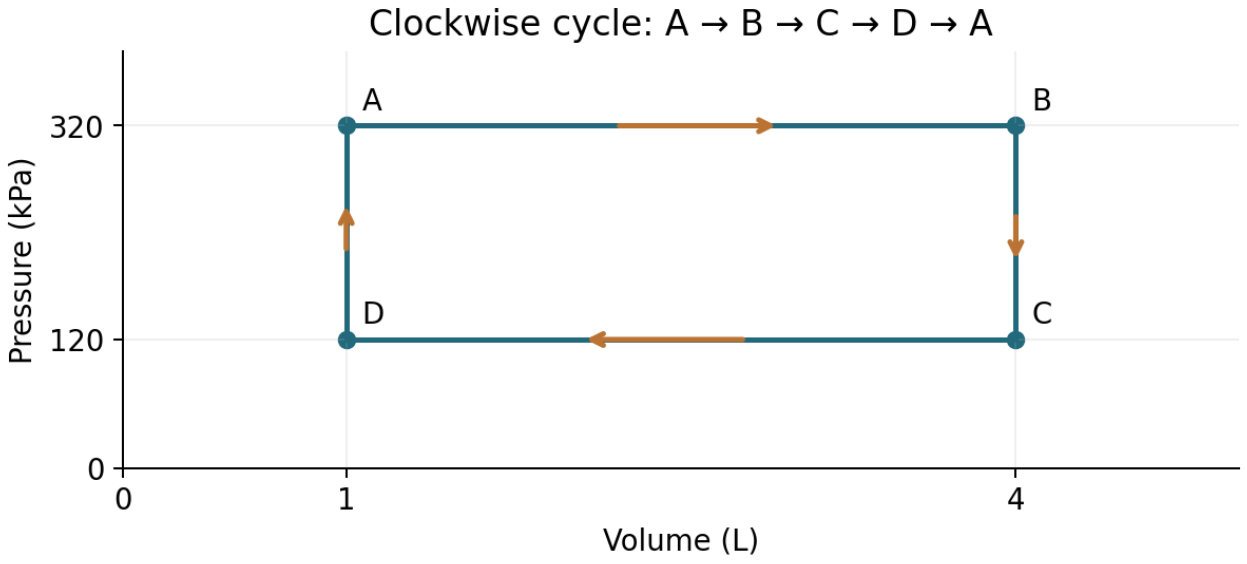
Use metric units and show reasoning. Suggested practice time: 45–55 minutes. These are instructional practice timings, not official event allocations. The full official event includes device testing. Use 273.15 for Celsius-to-Kelvin conversion and supplied constants. Q is positive into the system; W is positive by it; $\Delta U = Q - W$. Closed-system energy calculations neglect bulk kinetic/potential changes unless stated. No calculus, trigonometry, logarithms or State/National-only radiation/third-law calculations are required. All datasets are original illustrative models.

Tests 1–3 are focused practice and are not full official written-test simulations. Test 4 covers all five Regional written categories; device items use supplied foundational models and do not require deferred Unit 7.

Reference pages precede the question pages. Each question page has space for two responses. Use extra paper if needed and label every answer with its question ID. Answers and grading rubrics are in the separate instructor edition.

T03-F1 - Rectangular cycle

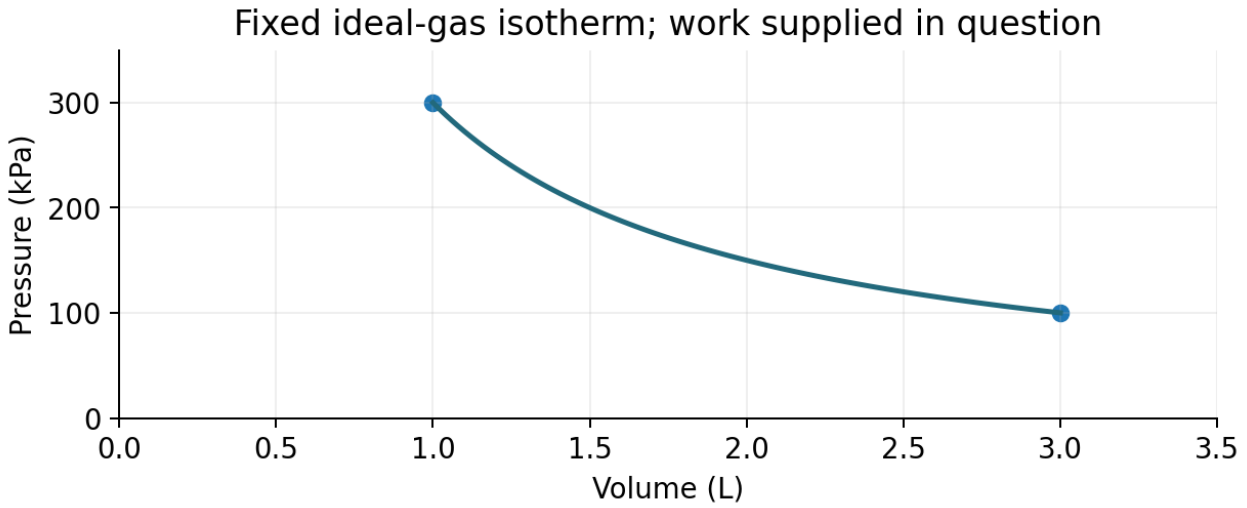
Slowly mechanically balanced paths. Follow A→B→C→D→A. 1 kPa·L=1 J.



State	V (L)	P (kPa)
A	1	320
B	4	320
C	4	120
D	1	120

T03-F2 - Isothermal path

PV=300 kPa·L for this fixed ideal-gas teaching model. Use the supplied work in Q24; do not integrate or replace the curve with a rectangle.



V (L)	P (kPa)
1	300
1.5	200
2	150
3	100

T03 – Questions 1–2

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q01 (2 points) Which statement describes internal energy U? Choose and explain.

A. Only energy presently crossing a wall

B. Microscopic energy of the system state

C. Work stored as a substance

D. Temperature in joules

T03-Q02 (2 points) For a rigid closed vessel, which statement about work is always justified by constant volume alone? Choose and explain.

A. No energy can enter

B. All work must be zero

C. PV boundary work is zero

D. Temperature must stay fixed

T03 – Questions 3–4

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q03 (2 points) Which condition defines an adiabatic process? Choose and explain what it does not necessarily imply.

A. No heat transfer

B. Constant temperature

C. Constant pressure

D. No work transfer

T03-Q04 (2 points) Which proposal contradicts the second law under ordinary positive-temperature conditions? Choose and explain.

A. A cup cools in a cooler room

B. A refrigerator uses work

C. Friction warms a brake

D. A cycle converts single-reservoir heat entirely into work with no other effect

T03 – Questions 5–6

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q05 (2 points) A closed system receives 620 J of heat and does 230 J of work. Find ΔU.

T03-Q06 (2 points) A fixed ideal gas expands isothermally from 90 kPa at 6.0 L to 9.0 L. Find its final absolute pressure.

T03 – Questions 7–8

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q07 (2 points) Convert 75 kPa·L to joules, explaining the conversion.

T03-Q08 (2 points) A fixed ideal gas at constant volume doubles its Kelvin temperature. How does absolute pressure change? Explain.

T03 – Questions 9–10

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q09 (2 points) A fixed ideal gas is heated at constant pressure from 250 K to 325 K. Its initial volume is 2.0 L. Find the final volume.

T03-Q10 (2 points) All affected parts of an isolated combination have supplied entropy changes −5 J/K and +8 J/K. Find the total and state whether the signs alone conflict with the second law.

T03 – Questions 11–12

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q11 (2 points) A working gas completes a thermodynamic cycle. What is its net internal-energy change? Can net work still be nonzero?

T03-Q12 (2 points) An insulated ideal-gas sample does 150 J of work on its surroundings. What is ΔU, and does its temperature rise or fall?

T03 – Questions 13–14

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q13 (3 points) A gas expands slowly against constant opposing pressure 140 kPa from 1.5 L to 4.0 L. Find its boundary work. If Q=510 J enters, find ΔU.

T03-Q14 (3 points) During compression, the surroundings do 280 J of work on a gas while the gas releases 95 J of heat. Find ΔU with the work-by convention and explain its sign.

T03 – Questions 15–16

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q15 (3 points) A fixed ideal gas is compressed isothermally. Work done by the gas is −210 J. Find ΔU and Q, and explain the direction of heat transfer.

T03-Q16 (3 points) An adiabatic compression supplies 540 J of work to a fixed ideal gas with supplied internal-energy coefficient C_U=90 J/K. Find ΔU and ΔT.

T03 – Questions 17–18

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q17 (3 points) Use the T03 rectangular cycle A→B→C→D→A. Find work on each horizontal segment and the net work. Assume slowly balanced pressure-volume paths.

T03-Q18 (3 points) Reverse the cycle in Q17. State the new net work and net heat into the gas over the complete cycle, explaining why its net ΔU remains zero.

T03 – Questions 19–20

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q19 (3 points) Three stages of a four-stage cycle have internal-energy changes +170 J, −260 J and +40 J. Find the fourth change. If that stage does 65 J of work, find its heat input.

T03-Q20 (3 points) An engine absorbs 1800 J and rejects 1080 J each cycle. Its reservoirs are 600 K and 330 K. Calculate work, actual efficiency and the modern ideal maximum.

T03 – Questions 21–22

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q21 (3 points) A student uses W/Q_net as heat-engine efficiency. Explain why this denominator is wrong and state what the ratio would give for a cycle with nonzero net output.

T03-Q22 (3 points) A gas returns to the same state every cycle, but friction is present. A student concludes each cycle is reversible because the gas's entropy returns to its initial value. Diagnose the error.

T03 – Questions 23–24

Name/team: _____

Show reasoning, equations and units. Use the reference pages as needed.

T03-Q23 (3 points) Two paths connect the same initial and final states of a closed system. Path A has Q=400 J and W=240 J. Path B does 90 J of work. Find Q for B and explain why it differs from A.

T03-Q24 (3 points) The T03 isotherm goes from 300 kPa at 1.0 L to 100 kPa at 3.0 L. The measured work for this ideal-gas expansion is supplied as 330 J. Find Q. Explain why using the starting pressure times the entire volume change is not a valid work calculation.